

DR ESLAM AHMED
eliteigcse.com

Sequences and Series

Write the first few terms. A recurrence builds each from the last; a sigma sum hides a known series; a periodic one repeats.

THE TOPIC ON ONE MAP

Sequences and Series

01 — Recurrence relations

$u_{n+1} = f(u_n)$ terms in k sum given $\rightarrow k$

02 — Sigma notation

write the first terms arithmetic or geometric? split the sum

03 — Periodic, increasing

order of a cycle sum of many cycles $(-1)^n$

WRITE THE FIRST FEW TERMS

$$\sum_{r=1}^n u_r = u_1 + u_2 + \cdots + u_n$$

number of terms = top – bottom + 1 · the pattern tells you: arithmetic, geometric, or periodic

01 Recurrence Relations: $u_{n+1} = f(u_n)$

Put $n = 1$ to get u_2 from u_1 , then $n = 2$ for u_3 . Keep the unknown constant as a letter; a sum or a given term then gives the equation for it.

WHEN YOU SEE → YOU DO

“find, in terms of k , an expression for u_3 ”

→ u_2 in k , then u_3 in k — expand

“given that $\sum_{n=1}^3 u_n = 117$ ”

→ $u_1 + u_2 + u_3 = 117$: an equation in k

“ $u_{n+2} = 3u_{n+1} - 2u_n$ ”

→ needs two starting terms: use the given ones

“ $u_{n+1} = (-1)^n u_n + k$ ”

→ $(-1)^n$ alternates the sign each step

“find the larger possible value”

→ solve the quadratic, test both

WORKED MODEL • OCTOBER 2021

$u_1 = 6$, $u_{n+1} = ku_n + 3$, $k > 0$, and $u_1 + u_2 + u_3 = 117$. Find k .

1 $u_2 = 6k + 3$, $u_3 = 6k^2 + 3k + 3$

generate

2 $6 + 6k + 3 + 6k^2 + 3k + 3 = 117 \Rightarrow 6k^2 + 9k - 105 = 0$

sum

ANSWER $2k^2 + 3k - 35 = 0 \Rightarrow k = \frac{7}{2} (k > 0)$

IDEA 01

Practise: Recurrence Relations: $u_{n+1} = f(u_n)$

My questions first, then real past papers.

Q1**DR ESLAM**

Generate terms

P2 · TOPIC 7

2 marks

$u_1 = 2$, $u_{n+1} = 3u_n - 1$. Find u_2 , u_3 and u_4 .

Q2**DR ESLAM**

Find the constant

P2 · TOPIC 7

4 marks

$u_1 = 4$, $u_{n+1} = ku_n + 2$, $k > 0$. Given $u_3 = 22$, find k .

Write u_2 and u_3 in k first.

PAST PAPERS

Q3 **PAST PAPER** Pearson IAL WMA12 · January 2021 · Q8**P2 · TOPIC 7** **7 marks**

8. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where p is a constant.

- (a) Find an expression for a_2 in terms of p , giving your answer in simplest form.

(1)

Given that $\sum_{n=1}^3 a_n = p + 15$

- (b) find the possible values of a_2

(6)**(Total 7 marks)**

Q4 **PAST PAPER** Pearson IAL WMA12 · October 2021 · Q2**P2 · TOPIC 7** **5 marks**

2. A sequence is defined by

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for u_3 (2)

Given that $\sum_{n=1}^3 u_n = 117$

- (b) find the value of k . (3)

(Total 5 marks)

Q5 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q11**P2 · TOPIC 7** **8 marks**

11. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where a and b are constants.

(a) Find, in terms of a and b ,

(i) u_2

(ii) u_3

(2)

Given

- $\sum_{n=1}^3 u_n = 153$

- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of u_2

(3)

(Total for Question 11 is 8 marks)

Q6**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q1

P2 · TOPIC 7 5 marks

1. The sequence $u_1, u_2, u_3, \dots$ satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

- (a) find the value of u_2 (2)

- (b) find $\sum_{r=1}^4 (u_r + 2r)$ (3)

(Total for Question 1 is 5 marks)

2. A sequence of numbers $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where k is a constant.

- (a) Show that $u_5 = 7$

(3)

Given that $\sum_{r=1}^4 u_r = 30$

- (b) find the value of k .

(2)

- (c) Hence find the value of $\sum_{r=1}^{150} u_r$

(2)

(Total for Question 2 is 7 marks)

02 Sigma Notation: Spot the Series

Write out the first two or three terms and the last one. A constant difference: arithmetic, use $\frac{n}{2}(a+l)$. A constant ratio: geometric. A mixed expression: split the sum into parts you know.

WHEN YOU SEE → YOU DO

$$\sum_{r=1}^k (31 - 6r)$$

→ arithmetic: $a = 25, d = -6$

$$\sum_{r=1}^{\infty} 6 \times 0.25^r$$

→ geometric: $a = 1.5, r = 0.25, S_{\infty}$

$$\sum_{n=1}^{59} \{5n + 3(-1)^n\}$$

→ split: $5 \sum n$ plus $3 \sum (-1)^n$

$$\sum_{r=5}^{k+2}$$

→ number of terms = top - bottom + 1

$$\text{"given } \sum_{r=1}^k \dots = 620004, \text{ show } k = \dots"$$

→ geometric sum = number, then logs

WORKED MODEL · JANUARY 2020

$$u_n = 5n + 3(-1)^n. \text{ Find } \sum_{n=1}^{59} u_n.$$

$$1 \quad 5 \sum_{n=1}^{59} n = 5 \times \frac{59 \times 60}{2} = 8850$$

arithmetic part

$$2 \quad 3(-1 + 1 - 1 + \dots - 1) = 3(-1) = -3$$

59 terms: one -1 left

$$\text{ANSWER} \quad 8850 - 3 = 8847$$

IDEA 02

Practise: Sigma Notation: Spot the Series

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Arithmetic inside

P2 · TOPIC 7 2 marks

Evaluate $\sum_{r=1}^{20} (3r + 2)$.

Q9 DR ESLAM Geometric inside

P2 · TOPIC 7 2 marks

Evaluate $\sum_{r=1}^8 3 \times 2^r$.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 7 4 marks**1 (a)** Calculate

$$\sum_{r=1}^5 2r + 1$$

(2 marks)

- (b)** The sum given in part [a] is an arithmetic series.
Write down the first term and the common difference.

(2 marks)

Q11 **PRACTICE** Practice · Hard

P2 · TOPIC 7 4 marks

1 Given that $\sum_{r=1}^k (31 - 6r) = -943$,

(i) Show that $(3k + 41)(k - 23) = 0$

(ii) Hence, find the value of k .

(4 marks)

Q12 **PRACTICE** Practice · Very Hard

P2 · TOPIC 7 4 marks

7 Prove that, for all $n \geq 1$,

$$\sum_{r=1}^n (2r)^2 - \sum_{r=1}^n (2r-1)^2 = \sum_{r=1}^{2n} r$$

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q8

P2 · TOPIC 7 7 marks

8. (i) An arithmetic series has first term a and common difference d .

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\} \quad (3)$$

- (ii) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5 (1)

(b) $\sum_{n=1}^{59} u_n$ (3)

(Total 7 marks)

Q14 PAST PAPER Pearson IAL WMA12 · June 2025 · Q7

P2 · TOPIC 7 9 marks

7: **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) A **geometric** series begins

$$10 + 8 + 6.4 + \dots$$

- (a) Find the sum to infinity of this series.

(2)

Given that the k th term of this series is less than 0.0005

- (b) use algebra to find the smallest possible value of k .

(3)

- (ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first n terms of this series is S_n
find the greatest possible value of S_n

(4)

(Total for Question 7 is 9 marks)

03 Periodic, Increasing and Decreasing Sequences

Generate terms until one repeats: the number of terms in one cycle is the order. For a long sum, add one cycle, multiply by the number of full cycles, and add the terms left over.

WHEN YOU SEE → YOU DO

“show that the sequence is periodic”

→ generate until u_1 comes back

“state the order”

→ the length of one cycle

“ $\sum_{n=1}^{70} u_n$ ” for order 3

→ 23 cycles + u_1

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

→ exact values repeat every 3

“periodic of period 4” with a sum given

→ 12 cycles + $u_1 + u_2 =$ the sum

“increasing, decreasing, or neither?”

→ compare u_{n+1} with u_n

WORKED MODEL • JANUARY 2024

$u_1 = 3, u_{n+1} = \frac{u_n - 3}{u_n - 2}$. Show it is periodic and find $\sum_{n=1}^{70} u_n$.

1 $u_2 = 0, u_3 = \frac{3}{2}, u_4 = 3$

repeats: order 3

2 one cycle: $3 + 0 + \frac{3}{2} = \frac{9}{2}$; $70 = 23 \times 3 + 1$

count cycles

ANSWER $23 \times \frac{9}{2} + 3 = \frac{213}{2}$

IDEA 03

Practise: Periodic, Increasing and Decreasing Sequences

My questions first, then selected practice, then real past papers.

Q15 DR ESLAM Periodic

P2 · TOPIC 7 4 marks

$u_1 = 2, u_{n+1} = \frac{1}{1 - u_n}$. Show that the sequence is periodic, state its order, and find $\sum_{r=1}^{30} u_r$.

SELECTED PRACTICE

Q16**PRACTICE**

Practice · Medium

P2 · TOPIC 7 6 marks

- 5 (a) The terms of a sequence are defined by $u_k = k^2$ for all $k \geq 1$.

State, with a reason, whether this sequence is increasing, decreasing, or neither.

(1 mark)

- (b) It can be shown that, for all $n \geq 1$,

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$$

Using that formula,

Calculate $\sum_{r=1}^{50} u_r$

(2 marks)

- (c) Find the value of $51^2 + 52^2 + 53^2 + \dots + 99^2 + 100^2$, i.e. the sum of the squares of all the integers between 51 and 100 inclusive.

(3 marks)

Q17**PRACTICE**

Practice · Hard

P2 · TOPIC 7 7 marks

- 5 (a) The terms of a sequence are defined, for all $k \geq 1$, by $u_k = (-1)^k \times k^2$.

State, with a reason, whether this sequence is increasing, decreasing, or neither.

(1 mark)

- (b) It can be shown that, for all $n \geq 1$,

$$\sum_{r=1}^n (2r)^2 = \frac{2n(n+1)(2n+1)}{3} \text{ and } \sum_{r=1}^n (2r-1)^2 = \frac{n(2n+1)(2n-1)}{3}$$

Using those formulas,

Show that $\sum_{r=1}^{100} u_r = \sum_{r=1}^{100} r$.

(6 marks)

Q18**PRACTICE**

Practice · Very Hard

P2 · TOPIC 7 5 marks

6 (a) A sequence is defined for all $k \geq 1$ by

$$u_k = -2k \times (\cos(k\pi))^{k+1}$$

Determine, giving reasons for your answer, whether the sequence is increasing, decreasing, or neither.

(3 marks)

(b) A different sequence is defined for all $k \geq 1$ by

$$v_k = \sin(kq\pi)$$

where q is a real constant.

Given that the sequence is not periodic,
suggest a possible value for q , giving a reason for your answer.

(2 marks)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA12 · October 2022 · Q3

P2 · TOPIC 7 7 marks

3. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i) a_1

(ii) a_2

(iii) a_3

(3)

(b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

(Total for Question 3 is 7 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2023 · Q2**P2 · TOPIC 7** **5 marks**

2. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

(a) Find the value of u_2 , the value of u_3 and the value of u_4

(3)

(b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

(Total for Question 2 is 5 marks)

Q21 PAST PAPER Pearson IAL WMA12 · January 2024 · Q5

P2 · TOPIC 7 8 marks

5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

(ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N}$$

(a) Show that this sequence is periodic. (2)(b) State the order of this sequence. (1)

(c) Hence find

$$\sum_{n=1}^{70} u_n$$

(2)

(Total for Question 5 is 8 marks)

Q22 **PAST PAPER** Pearson IAL WMA12 · June 2026 · Q3**P2 · TOPIC 7** **9 marks**

3. The first four terms of a sequence are

$$3, -6, u_3, -24, \dots$$

(a) Decide, showing detailed reasoning, whether the sequence could be

- (i) an arithmetic sequence,
- (ii) a geometric sequence.

State, for each part, the value of u_3 if such a sequence is possible.

(6)

(b) Given instead that

- the sequence is periodic of period 4
- $\sum_{i=1}^{50} u_i = 141$

find the value of u_3

(3)

(Total for Question 3 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------|---|
| 01 | Recurrence | Generate in k ; the sum gives the equation. |
| 02 | Sigma | Write the terms; arithmetic, geometric or split. |
| 03 | Periodic | One cycle; cycles $\times$ cycle-sum + leftovers. |

BOTTOM LINE Always write the first few terms $\rightarrow$ the pattern tells you which formula to use.

Answers

Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · RECURRENCE RELATIONS: $u_{n+1} = f(u_n)$

- Q1** Dr Eslam · Generate terms 5, 14, 41
- Q2** Dr Eslam · Find the constant $u_3 = 4k^2 + 2k + 2 = 22 \Rightarrow (2k + 5)(k - 2) = 0$:
 $k = 2$
- Q3** Pearson IAL WMA12 · January 2021 · Q8 $(a) a_2 = 2p^2 - 7. \quad (b) a_2 = -7 \text{ or } \frac{1}{2}$
- Q4** Pearson IAL WMA12 · October 2021 · Q2 $(a) u_3 = 6k^2 + 3k + 3. \quad (b) k = \frac{7}{2}$
- Q5** Pearson IAL WMA12 · June 2023 · Q11 $(a)(i) u_2 = b - 3a. \quad (a)(ii) u_3 = b - ab + 3a^2.$
 $(b) \text{ Proven.} \quad (c) 21$
- Q6** Pearson IAL WMA12 · June (R paper) 2024 · Q1 $(a) u_2 = 6. \quad (b) 37$
- Q7** Pearson IAL WMA12 · October 2024 · Q2 $(a) \text{ Proven.} \quad (b) k = 15. \quad (c) 1125$

IDEA 02 · SIGMA NOTATION: SPOT THE SERIES

- Q8** Dr Eslam · Arithmetic inside $a = 5, l = 62: 10(5 + 62) = 670$
- Q9** Dr Eslam · Geometric inside $a = 6, r = 2: \frac{6(2^8 - 1)}{1} = 1530$
- Q10** Practice · Easy (a) $3 + 5 + 7 + 9 + 11 = 35$ (b) $a = 3, d = 2$
- Q11** Practice · Hard (i) $31k - 3k(k + 1) = -943 \Rightarrow 3k^2 - 28k - 943 = 0$
(ii) $k = 23$
- Q12** Practice · Very Hard $\sum\{(2r)^2 - (2r - 1)^2\} = \sum(4r - 1) = 2n^2 + n = \frac{2n(2n+1)}{2} = \sum_{r=1}^{2n} r$
- Q13** Pearson IAL WMA12 · January 2020 · Q8 (i) Proven. (ii)(a) 22. (ii)(b) 8847
- Q14** Pearson IAL WMA12 · June 2025 · Q7 (i)(a) 50. (i)(b) 46. (ii) 52033

IDEA 03 · PERIODIC, INCREASING AND DECREASING SEQUENCES

- Q15** Dr Eslam · Periodic $2, -1, \frac{1}{2}, 2, \dots$: order 3; $10 \times 1.5 = 15$
- Q16** Practice · Medium (a) increasing: k^2 grows for $k \geq 1$ (b) 42925
(c) $338350 - 42925 = 295425$
- Q17** Practice · Hard (a) neither: the sign alternates (b) $171700 - 166650 = 5050 = \sum_{r=1}^{100} r$
- Q18** Practice · Very Hard (a) $(\cos k\pi)^{k+1} = (-1)^{k(k+1)} = 1$, so $u_k = -2k$: decreasing (b) any irrational q , e.g. $\sqrt{2}$: kq is never a whole number of periods
- Q19** Pearson IAL WMA12 · October 2022 · Q3 (a) $a_1 = \frac{1}{4}, a_2 = \frac{1}{4}, a_3 = 1$. (b) $\frac{2599}{2}$
- Q20** Pearson IAL WMA12 · October 2023 · Q2 (a) $u_2 = \frac{2}{3}, u_3 = -4, u_4 = 3$. (b) -8
- Q21** Pearson IAL WMA12 · January 2024 · Q5 (i) 2. (ii)(a) Periodic. (ii)(b) Order 3.
(ii)(c) $\frac{213}{2}$
- Q22** Pearson IAL WMA12 · June 2026 · Q3 (a)(i) $d = -9$ and $3 + 3(-9) = -24$: yes, $u_3 = -15$
(ii) $r = -2$ and $3(-2)^3 = -24$: yes, $u_3 = 12$ (b)
 $12(u_3 - 27) + 3 - 6 = 141$: $u_3 = 39$